T. H. MIHIRAN THILAKARATHNA

Aspiring AI Automation & Python Developer | IT Undergraduate

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PROFESSIONAL SUMMARY

Final-year Information Technology undergraduate with a strong interest in AI, automation, and software development. Experienced in building full-stack applications using Python, FastAPI, React, Next.js, and Node.js, with hands-on experience in designing REST APIs and developing AI-powered solutions using LLM APIs, TensorFlow, and PyTorch. Passionate about leveraging AI to solve real-world problems, automate business processes, and build scalable, intelligent applications. A quick learner with strong problem-solving skills, eager to contribute to an AI Automation team while continuously expanding expertise in modern AI technologies.

EDUCATION

BSc. Information Technology | Final-Year Undergraduate **2023 – Present**

Rajarata University of Sri Lanka, Mihintale

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, SQL
AI / LLMs: LLM APIs (Google Gemini), Prompt Engineering, TensorFlow, PyTorch, Scikit-learn, NLP, Computer Vision
Backend & APIs: FastAPI, Node.js, Express.js, REST API design, Firebase (Auth & Cloud Firestore)
Databases: MongoDB, MySQL, SQLite
Web & Mobile: React, Next.js, Kotlin (Jetpack Compose), Flutter
Tools & Platforms: Git, VS Code, Postman, Vercel, Render, Hugging Face Spaces, Jupyter/Colab
Currently Learning: LangChain, Retrieval-Augmented Generation (RAG), Vector Databases, Docker, n8n

PROJECTS

OmniModel Converter Live Demo | GitHub

- Full-stack ML model conversion platform with a FastAPI backend that automates model translation across PyTorch, TensorFlow, ONNX, and TFLite.
- Designed clean, well-documented REST API contracts for programmatic model conversion; deployed on Vercel and Hugging Face Spaces.

Tech: Python, FastAPI, PyTorch, TensorFlow, ONNX, TFLite, Next.js

Lumo AI Assistant GitHub

- AI-powered assistant integrating the Google Gemini LLM API for natural-language conversations and automated PDF document processing.
- Built backend data handling with Firebase Auth and Cloud Firestore, and designed prompt flows for reliable, context-aware responses.

Tech: Kotlin, Google Gemini API, Firebase, PDFBox-Android

Vendora — SaaS E-Commerce Admin Dashboard Live Demo | GitHub

- Full-stack admin dashboard with role-based access control, product/order management, and real-time notifications.
- Built REST API endpoints with Node.js/Express.js and a reusable React component library; backend deployed on Render.

Tech: React, Node.js, Express.js, MongoDB, Tailwind CSS

Facial Occlusion Classification for Restricted Environments GitHub

- MobileNetV2 transfer learning model achieving 98% test accuracy classifying faces as clear, helmet-occluded, or mask-occluded.

Tech: Python, TensorFlow, Keras, MobileNetV2, Scikit-learn

MoodTune — Emotion-Aware Music Recommender GitHub

- Android app using a custom CNN (FER-2013 dataset) via TensorFlow Lite and Google ML Kit for real-time facial emotion detection with mood-matched music recommendations.

Tech: Java, TensorFlow Lite, Google ML Kit, Python/Keras, CameraX, MVVM

WORK & LEADERSHIP EXPERIENCE

Batch Coordinator **2026 – Present**

Department of Information Technology, Rajarata University of Sri Lanka

- Coordinated communication between students and academic staff while managing announcements and digital information sharing.

Trainee **Dec 2022 – Jul 2023**

People's Bank, Avissawella Branch

- Assisted with daily banking operations, financial documentation, and customer service in a regulated environment, building attention-to-detail and professional communication skills.

ACHIEVEMENTS & AWARDS

Grand Finalist — InnovateX 2025 **2025**

- Selected as a Grand Finalist at InnovateX 2025 (Zebra Technologies, Galle Face Hotel), one of Sri Lanka's premier technology innovation competitions, representing Rajarata University as Team LoopCode.